2022 IEEE International Conference on Robotics and Automation (ICRA)
May 23-27, 2022. Philadelphia, PA, USA

Constrained Variable Impedance Control using Quadratic Programming

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Abstract— **This paper proposes a quadratic programming (QP)-based variable impedance control (VIC) algorithm to solve contact-rich trajectory tracking problems with impedance, position and velocity constraints. To the best of our knowledge, the impedance constraints which are significant to ensure the worst contact compliance have never been considered in other previous works. To handle the impedance constraints of the VIC algorithm, a novel impedance model where the impedance parameters are directly served as the control input is established. The impedance-constrained VIC design problem is then formulated as a QP problem which can be efficiently solved. To handle the position and velocity constraints, a complementary force is introduced into the novel impedance model. The complementary force will appear to prevent the constraints violation when the robot approaches the constrained area. The design problem of the complementary force is also transformed into a QP problem. Combing these two QP solutions, the VIC algorithm with both impedance, position and velocity constraints can be obtained. Finally, various experiments are conducted to show the effectiveness of the proposed QP-based constrained VIC algorithm.**

I. INTRODUCTION

Impedance control is a reliable control technique for contact-rich robotic tasks [1]. The commonly used impedance control is the constant impedance control (CIC) where impedance parameters are fixed and determined by humans according to their domain experiences. However, CIC is only suitable for simple or structured tasks due to its single control mode. Variable impedance control (VIC), i.e., impedance control with variable impedance parameters, are more effective to handle complex contact-rich tasks, such as robotic assembly [2], human-robot interactions [3], robotic rehabilitation [4] and robotic surgery [5], etc. Thus, various works [6], [7] have been done to investigate VIC algorithms in recent years.

The most intuitive VIC algorithms may be heuristic-based VIC algorithms of which the variable impedance laws are directly designed according to general knowledge or rules. In [8], the robot stiffness was set as the same value as the estimated human stiffness when executing an objective-transporting task. In [9], [10], the robot damping was determined according to the estimated human intentions to save the human-cost energy. In [11], the robot damping was online tuned based on the change of the interaction force to conduct safe, compliant motions. The above approaches are usually effective for simple tasks without quantitative or accuracy requirements, such as most human-robot cooperation tasks.

This paper mainly focuses on optimal VIC algorithms of which the variable impedance laws are often numerically computed using optimization techniques. Recently, reinforcement learning (RL) algorithms are more and more popular to design variable impedance laws. By designing the reward functions to be consistent with the task objectives, RL algorithms can generate useful variable laws. According to the complexities of tasks, different RL algorithms can be used to generate variable impedance laws, such as classical RL algorithm used in [12], deep RL algorithms used in [13], [14]. One of the main issues for these RL-based methods is that all of the variable impedance laws were obtained by trial-and-errors which are time-consuming and unsafe for robotic applications. Thus, model-based optimization methods may be more suitable to design variable impedance laws, especially in contact-rich tasks.

In [15]–[17] where trajectory-tracking tasks were considered, approximated models were established and model predictive control (MPC) approaches were presented to design variable impedance laws in order to consider the tracking accuracy and contact compliance simultaneously. In [18], [19], LQR and RL approaches were used to derive variable impedance laws where the former and latter were utilized to handle the model-based design and model-free design, respectively. A common issue for all of the previously mentioned approaches is that they did not consider the practical constraints of robots or tasks, such as position and velocity constraints. A most relevant work which handled these practical constraints under the impedance control framework was presented in [20] by combining the CIC and MPC approaches. To the best of our knowledge, no works have been done to design the variable impedance laws when simultaneously considering these practical constraints. Moreover, the variable range of the impedance parameters, termed as impedance constraints which are significant to ensure the worst contact compliance, are also never considered in previous VIC researches.

This paper presents a quadratic programming (QP)-based VIC algorithm to solve contact-rich trajectory tracking tasks with both impedance, position, and velocity constraints. The main contributions are listed as follows:

(1) The impedance constraints which are significant to ensure the worst contact compliance is originally considered in this paper, and we transform the design problem of the

This work was supported by the National Natural Science Foundation of China under Grant 61973275, the Zhejiang Provincial Natural Science Foundation of China under Grant LD21F030002, and the Fundamental Research Funds for the Provincial Universities of Zhejiang RF-A2020004. (Corresponding author: Andong Liu.)

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impedance-constrained VIC algorithm into a QP problem based on a novel presented impedance model.

(2) We present a complementary force for the novel impedance model to handle the position and velocity constraints. The design problem of the complementary force is also formulated as a QP problem.

(3) By combining the solutions of the two QP problems, a VIC algorithm with both impedance, position and velocity constraints is proposed.

Finally, the necessity of setting the impedance constraints are validated by a collision experiment, and various experiments have been done to show the effectiveness of the presented QP-based constrained VIC algorithm.

The remaining contents are organized as follows: The control background of the robot dynamic control and impedance control are briefly introduced in Section II. The studied problem of this paper is stated in Section III. The proposed algorithm is introduced in Section IV. Experiments are conducted in Section V. A conclusion is given in Section VI.

II. Control background

A. Robot dynamic control

Consider an n-DOF robot dynamic model as follows

$$H(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = J^T f + \tau \tag{1}$$

where $H(q) \in R^{n\times n}$, $C(q,\dot{q}) \in R^{n\times n}$ and $g(q) \in R^n$ are the inertia matrix, Coriolis/Centrifugal matrix, and gravity vector, respectively. The matrix $J(q) \in R^{m\times n}$ is the robot Jacobian and the vector $f \in R^m$ is the interaction force. The control objective is to track the desired acceleration υ in workspace. The relationship between the workspace acceleration $\ddot{\xi} = \begin{bmatrix}\ddot{x} & \ddot{y} & \ddot{z} & \ddot{\theta}_x & \ddot{\theta}_y & \ddot{\theta}_z\end{bmatrix}^T$ and joint-space acceleration $\ddot{q}$ can be derived as

$$\ddot{q} = J^+(\ddot{\xi} - \dot{J}\dot{q}) \tag{2}$$

where J^+ is the pseudo-inverse of the Jacobian matrix J. Thus, the controller can be designed as

$$\tau = H(q)J^+(\upsilon - \dot{J}\dot{q}) + C(q,\dot{q})\dot{q} + g(q) - J^T f \tag{3}$$

B. Impedance control

The impedance control in work space can be described as

$$M\ddot{\xi} + D(\dot{\xi} - \dot{\xi}_r) + K(\xi - \xi_r) = f \tag{4}$$

where M, D and K are inertia, damping and stiffness matrices. Assume M, D and K are positive definite and diagonal, we can decouple the model (4) to several one-dimensional models. As an example, the one-dimensional model in x-dimension is

$$m_x\ddot{x} + d_x(\dot{x} - \dot{x}_r) + k_x(x - x_r) = f_x \tag{5}$$

where the subscript x indicates the considered dimension, x_r represents the reference position for x-dimension. Since every dimension of the state has the same control strategy, the subscripts are omitted for convenience and we take the x-dimension as the example in the rest of the paper. After omitting subscript x, we can rewrite the model (5) as

$$\begin{cases} \dot{s} = As + b\upsilon \\ s = \begin{bmatrix} x & \dot{x} & f \end{bmatrix}^T \\ A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ b = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T \\ \upsilon = \begin{bmatrix} \frac{-k}{m} & \frac{-d}{m} & \frac{1}{m} \end{bmatrix}^T \tilde{s} \\ \tilde{s} = s - s_r = \begin{bmatrix} x - x_r & \dot{x} - \dot{x}_r & f \end{bmatrix}^T \end{cases} \tag{6}$$

Remark 1: It should be noted that $\upsilon = \ddot{x}$ is the desired workspace acceleration that needs to be submitted to the controller (3).

The model (6) assumes $\dot{f} = 0$, which may raise the gap between the established model and the real model. Although accurately modeling $\dot{f}$ can increase the control performance, it goes far beyond the scope of this paper. By determining the impedance parameters, i.e., m, d and k, υ is actually served as a state-feedback control law for the impedance model (6).

III. Problem statement

Impedance parameters m, d and k have distinct physical meanings and they actually determine the property of an inertia-damping-spring system. For example, large stiffness k or damping d will decrease the tracking error $x - x_r$ or $\dot{x} - \dot{x}_r$ while making the system in-compliant with the interaction force f, which may not suitable for physical interaction tasks such as human-robot interaction. On the contrary, too small stiffness k or damping d results a compliant system but raises a large tracking error, which may violate the objective of the task, e.g., contact-rich trajectory tracking. Thus, how to determine the optimal impedance parameters to both consider the contact-compliance and tracking error is important for impedance control.

Numerous VIC methods have been presented to adaptively tune the impedance parameters online according to the model (6), such as MPC [15]–[17] and LQR-based [18], [19] approaches. However, they never consider the range of impedance parameters which is termed as impedance constraints in this paper. Impedance constraints are significant to ensure some extreme contact properties, for example, the maximum value of stiffness k can determine the worst contact compliance.

The last issue is about the position and velocity constraints which are common but significant for robotic tasks, such as robotic surgery, robotic rehabilitation, etc. Obviously, directly using the state-feedback control law as υ in (6) can not ensure the position and velocity constraints.

As a conclusion, the considered problems of this paper are listed as

(1) Designing an optimal VIC algorithm under impedance constraints.

(2) Ensuring the position and velocity constraints for the VIC algorithm.

IV. METHODOLOGY

This section is divided into two parts to solve the two considered problems, respectively, by using QP techniques.

A. QP for VIC with Impedance Constraints

It is difficult to constrain the impedance parameters using the model (6) since the impedance parameters are hidden in the control input. To handle this issue, we define a new input $u = \begin{bmatrix} \frac{-k}{m} & \frac{-d}{m} & \frac{1}{m} \end{bmatrix}$, and the model (6) can be rewritten as

$$\dot{s} = As + b\tilde{s}^T u \tag{7}$$

By discretizing the above model, one has

$$s_{t+1} = (AT + I)s_t + T(b + \beta)\tilde{s}_t^T u_t \tag{8}$$

where $\beta = \begin{bmatrix} \frac{T}{2} & 0 & 0 \end{bmatrix}^T$.

Remark 2: The uniform-acceleration model is used here to discretize the model, and thus there will appear an additional vector β.

The input u_t is designed to solve the following optimization problem

$$\min_{u_t} J(u_t) = \tilde{s}_{t+1}^T Q \tilde{s}_{t+1} + u_t^T R u_t \tag{9}$$

subject to

$$\begin{cases} m = c_m \\ d = 2\sqrt{m}\frac{k}{\sqrt{k_{min}}} \\ k \in [k_{min}, k_{max}] \end{cases} \tag{10}$$

where Q and R are positive semi-definite and positive definite matrices, respectively. The objective function (9) implies a trade-off between the tracking accuracy and the contact compliance. Constraints (10) raise a constant inertia, over-damping and variable stiffness system. The optimization problem (9), (10) can be rewritten as the following QP problem

$$\min_{u_t} J(u_t) = \frac{1}{2} u_t^T \overline{Q}_t u_t + \overline{p}_t^T u_t + \overline{c}_t \tag{11}$$

subject to

$$\begin{cases} \overline{G} u_t \preceq \overline{h} \\ \overline{A} u_t = \overline{b} \end{cases} \tag{12}$$

where c_t is independent with u_t, and

$$\begin{cases} \overline{Q}_t = 2T^2 \tilde{s}_t (b+\beta)^T Q (b+\beta) \tilde{s}_t^T + 2R \\ \overline{p}_t = 2T \tilde{s}_t (b+\beta)^T Q \left((AT+I)\, s_t - s_{r,t+1} \right) \\ \overline{G} = \begin{bmatrix} -c_m & 0 & 0 \\ c_m & 0 & 0 \end{bmatrix} \\ \overline{h} = \begin{bmatrix} k_{max} & -k_{min} \end{bmatrix}^T \\ \overline{A} = \begin{bmatrix} 0 & 0 & c_m \\ \frac{2\sqrt{c_m}}{\sqrt{k_{min}}} & -1 & 0 \end{bmatrix} \\ \overline{b} = \begin{bmatrix} 1 & 0 \end{bmatrix}^T \end{cases} \tag{13}$$

Remark 3: The QP constraint form (12) can represent more flexible impedance constraints such as $m \in [m_{min}, m_{max}]$, $d \in [d_{min}, d_{max}]$, $k \in [k_{min}, k_{max}]$. We here choose constraints (10) for more safe contact.

B. QP for VIC with Position and Velocity Constraints

It should be noted that with system (8), position constraints are difficult to be ensured since the damping force will disappear when the velocity error is zero. Thus, we introduce a complementary force f_c into the system (8) as

$$s_{t+1} = (AT + I)s_t + T(b+\beta)(\tilde{s}_t^T u^* + \frac{f_{c,t}}{m}) \tag{14}$$

where u^* is determined by solving the QP problem (11), (12). The model (14) can be rewritten as the following linear model

$$s_{t+1} = \overline{A} s_t + \overline{B} f_{c,t} - \overline{C} s_{r,t} \tag{15}$$

where

$$\begin{cases} \overline{A} = AT + I + T(b+\beta) u^{*T} \\ \overline{B} = \frac{1}{m} T(b+\beta) \\ \overline{C} = T(b+\beta) u^{*T} \end{cases} \tag{16}$$

To constrain the position and velocity, we design a complementary force sequence $\mathbf{f}_{c,t} = \begin{bmatrix} f_{c,t} & \cdots & f_{c,t+H-1} \end{bmatrix}^T$ by solving the following optimization problem

$$\min_{\mathbf{f}_{c,t}} J(\mathbf{f}_{c,t}) = \sum_{i=t}^{t+H-1} \frac{1}{2} \omega_i f_{c,i}^2 \tag{17}$$

subject to

$$\begin{cases} s_{i+1} = \overline{A} s_i + \overline{B} f_{c,i} - \overline{C} s_{r,i} \\ x_{i+1} \in [x_{min,i+1}, x_{max,i+1}] \\ \dot{x}_{i+1} \in [\dot{x}_{min,i+1}, \dot{x}_{max,i+1}] \end{cases} \tag{18}$$

where $\omega_{i,i=1,\cdots,t+H-1}$ are positive weight parameters. The objective function (17) implies that the original system (8) is preferable, that is, we want moderate the original system (8) as less as possible. By defining

$$\begin{cases} \mathcal{A} = \begin{bmatrix} \overline{A} \\ \overline{A}^2 \\ \vdots \\ \overline{A}^H \end{bmatrix} \\ \mathcal{B} = \begin{bmatrix} \overline{B} & & & \\ \overline{AB} & \overline{B} & & \\ \vdots & \vdots & \ddots & \\ \overline{A}^{H-1}\overline{B} & \overline{A}^{H-2}\overline{B} & \cdots & \overline{B} \end{bmatrix} \\ \mathcal{C} = \begin{bmatrix} \overline{C} & & & \\ \overline{AC} & \overline{C} & & \\ \vdots & \vdots & \ddots & \\ \overline{A}^{H-1}\overline{C} & \overline{A}^{H-2}\overline{C} & \cdots & \overline{C} \end{bmatrix} \end{cases} \tag{19}$$

and

$$\begin{cases} \mathbf{s}_{r,t} = \begin{bmatrix} s_{r,t}^T & s_{r,t+1}^T & \cdots & s_{r,t+H-1}^T \end{bmatrix}^T \\ \mathbf{x}_{min,t+1} = \begin{bmatrix} x_{min,t+1}^T & x_{min,t+2}^T & \cdots & x_{min,t+H}^T \end{bmatrix}^T \\ \mathbf{x}_{max,t+1} = \begin{bmatrix} x_{max,t+1}^T & x_{max,t+2}^T & \cdots & x_{max,t+H}^T \end{bmatrix}^T \\ \dot{\mathbf{x}}_{min,t+1} = \begin{bmatrix} \dot{x}_{min,t+1}^T & \dot{x}_{min,t+2}^T & \cdots & \dot{x}_{min,t+H}^T \end{bmatrix}^T \\ \dot{\mathbf{x}}_{max,t+1} = \begin{bmatrix} \dot{x}_{max,t+1}^T & \dot{x}_{max,t+2}^T & \cdots & \dot{x}_{max,t+H}^T \end{bmatrix}^T \end{cases} \tag{20}$$

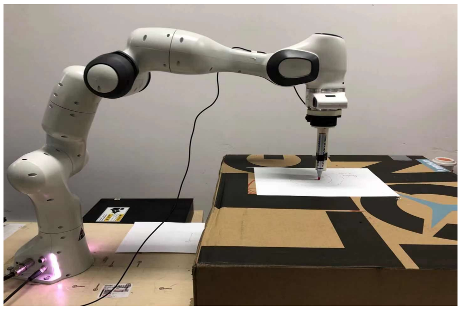

Fig. 1. Experimental task.

The optimization problem (17), (18) can be rewritten as the following QP problem

$$\min_{\mathbf{f}_{c,t}} J(\mathbf{f}_{c,t}) = \frac{1}{2}\mathbf{f}_{c,t}^T W \mathbf{f}_{c,t} \tag{21}$$

subject to

$$\overline{G}\mathbf{f}_{c,t} \preceq \overline{h} \tag{22}$$

where

$$\begin{cases} W = diag\{\omega_t, \cdots, \omega_{t+H-1}\} \\ \overline{G} = \begin{bmatrix} -\Psi \\ \Psi \\ -\Phi \\ \Phi \end{bmatrix} \mathcal{B} \\ \overline{h} = \begin{bmatrix} -\mathbf{x}_{min,t+1} \\ \mathbf{x}_{max,t+1} \\ -\dot{\mathbf{x}}_{min,t+1} \\ \dot{\mathbf{x}}_{max,t+1} \end{bmatrix} - \begin{bmatrix} -\Psi \\ \Psi \\ -\Phi \\ \Phi \end{bmatrix} (\mathcal{A}s_t - \mathcal{C}\mathbf{s}_{r,t}) \end{cases} \tag{23}$$

and

$$\begin{cases} \Psi = diag\{\beta^T, \cdots, \beta^T\} \\ \Phi = diag\{\gamma^T, \cdots, \gamma^T\} \\ \beta = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T \\ \gamma = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T \end{cases} \tag{24}$$

Concluded by the above analysis, the QP-based constrained VIC algorithm is given as Algorithm 1.

Algorithm 1 QP-based constrained VIC algorithm

for $t = 1, 2, \cdots$ **do**
  Observing the the state s_t
  Computing u^* by solving the QP problem (11), (12)
  Computing $\mathbf{f}_{c,t}^*$ by solving the QP problem (21), (22)
  Computing the desired acceleration $v_t = \tilde{s}_t^T u^* + \frac{f_{c,t}^*}{m}$
  Computing the control torque τ_t by using (3)
  Sending the control torque τ_t to the robot
end for

V. Experimental validation

A. Experiment setup

The experimental task is show in Fig. 1 where the robot is asked to track reference trajectories. In this paper, the rectangle and spiral trajectories are considered where the first one can validate the proposed methods more intuitively and the more complex spiral trajectory is used to further show the effectiveness of the method.

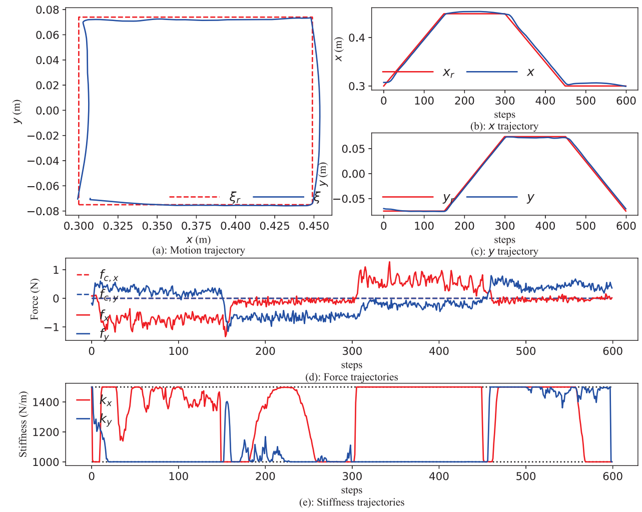

Fig. 2. Experimental results for rectangle-trajectory tracking with only impedance constraints. (a): Reference and real motion trajectories. (b): Reference and real x-direction trajectories. (c): Reference and real y-direction trajectories. (d): Force trajectories. (e): Stiffness trajectories.

In rectangle-trajectory tracking experiments, the robot firstly conducts the tracking task with only impedance constraints, and we show the advantage of impedance constraints when the robot contacts with an obstacle. Then, the robot tracks the rectangle trajectory with both impedance and position constraints. In this part we show that even there is an external force which pushes the robot to violate the position constraint, the robot will generate a resistance force to weaken the violation. In spiral-trajectory tracking experiments, impedance and position constrained tracking tasks are conducted to further show the performance of the proposed algorithm in complex scenes.

All experiments are conducted on the Franka-Panda robot and a computer with the Intel Core i7-8700K CPU. The proposed method and low level controller (3) are conducted in 50 Hz and 1000 Hz, respectively. We also provide a video for these experiments in attached files.

B. Experimental results

The rectangle-trajectory tracking experiment with only impedance constraints is firstly conducted, and the QP parameters in Section IV A are listed as $c_m = 1$, $k_{max} = 1500N/m$, $k_{min} = 1000N/m$ and

$$\begin{cases} Q = diag\{50, 0, 0\} \\ R = diag\{10^{-9}, 10^{-9}, 10^{-9}\} \end{cases} \tag{25}$$

The experimental result is shown in Fig. 2 from which we can see that the robot can tracks the reference trajectory well. The small tracking error is caused by the model error in low level controller (3) and the friction f_x, f_y shown in Fig. 2 (d). When the tracking error is large, the large stiffness will appear to increase the tracking accuracy. When the tracking

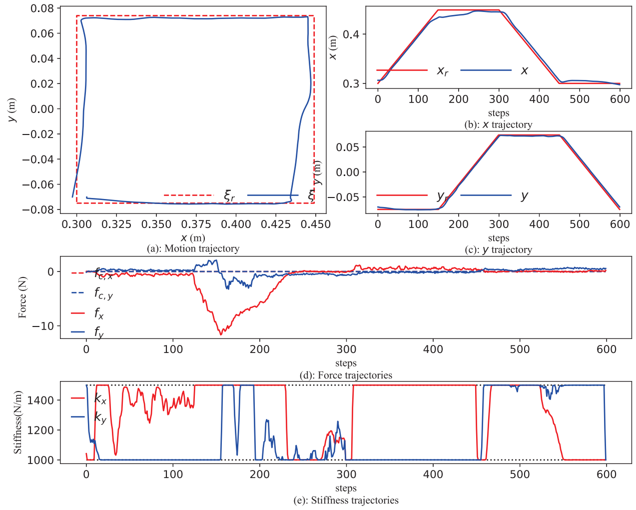

Fig. 3. Experimental results for rectangle-trajectory tracking with only impedance constraints under collision scene.

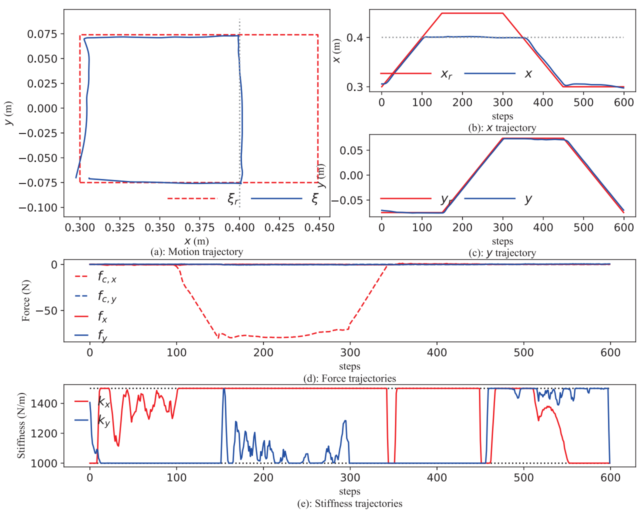

Fig. 4. Experimental results for rectangle-trajectory tracking with both impedance and position constraints.

error is small, the method generate small stiffness to save energy. From Fig. 2 (e) we can also find that the stiffness is limited as the constraints (12) show. To state the necessity of the impedance constraints, we further conduct an experiment where the robot will contact with an obstacle (a human hand), and the result is shown in Fig. 3. The contact appears at the low-right corner of the Fig. 3 (a), and its corresponding step is step 125. From Fig. 3 (e) we can find that the stiffness will rapidly increase since there will raise the large tracking error when the robot is stopped by the obstacle. If we do not set a constraint for the stiffness, there will raise a large contact force which may raise safety problem.

The experiment for rectangle-trajectory tracking with both impedance and position constraints is then conducted. QP parameters in Section IV B are given as $H = 5$, $W = 1$ and

$$x_{max,t+1} = x_{max,t+2} = \cdots = x_{max,t+H} = 0.4m \tag{26}$$

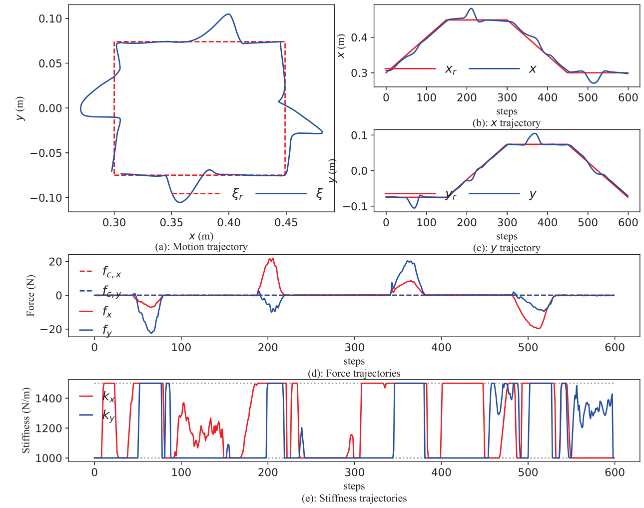

Fig. 5. Experimental results for rectangle-trajectory tracking with only impedance constraints under the violated force scene.

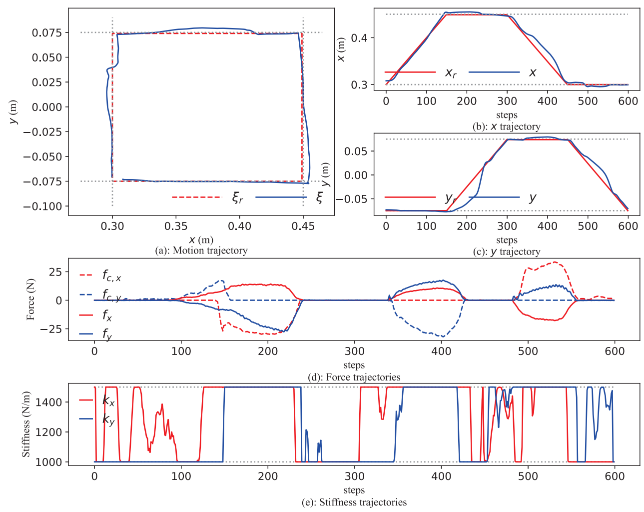

Fig. 6. Experimental results for rectangle-trajectory tracking with both impedance and position constraints under the violated force scene.

It should be noted that we only limit the x_{max} in this experiment for convenience. The result is show in Fig. 4. We can find from Fig. 4 (a), (b), (e) that the impedance and position constraints are both ensured. When the robot approaches the constrained position $x = 0.4m$, the complementary force $f_{c,x}$ will be generated to slow down and stop the robot, which is reflected by the Fig. 4 (d). We also provide more experiments with different position constraints in the attached video. To further show the effectiveness of the position constraints, we apply forces on the robot to violate the position constraints. The experimental results are shown in Fig. 5 and Fig. 6 where the former is only with the impedance constraints and the later is with both impedance and position constraints. From Fig. 5 we can find that without the complementary forces f_x and f_y, the real motion can be easily influenced by the external forces. However, if we set position constraints at the edges of the rectangle as the case

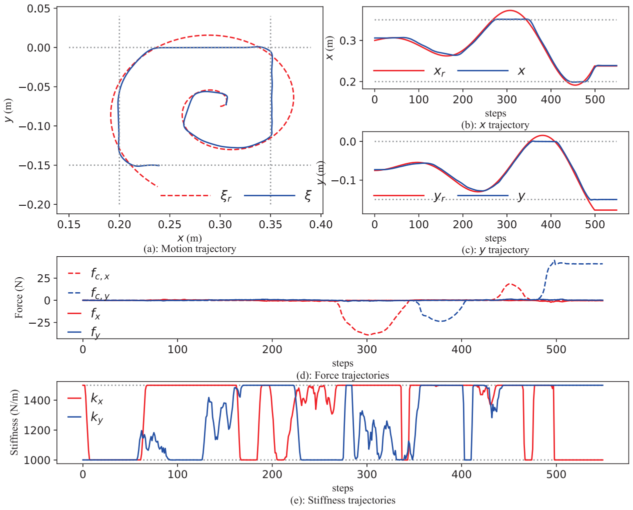

Fig. 7. Experimental results for spiral-trajectory tracking with both impedance and position constraints.

shown in Fig. 6, the external forces those will violate the position constraints will be neutralized by the complementary forces and thus the degree of the violation will be decreased.

To further show the effectiveness of the proposed method, the more complex spiral-trajectory tracking experiments are conducted. the robot is asked to track the trajectory with both impedance and position constraints. Here, the position constraints are given as follows

$$\begin{cases} x_{min,t+1} = x_{min,t+2} = \cdots = x_{min,t+H} = 0.2m \\ x_{max,t+1} = x_{max,t+2} = \cdots = x_{max,t+H} = 0.35m \\ y_{min,t+1} = y_{min,t+2} = \cdots = y_{min,t+H} = -0.15m \\ y_{max,t+1} = y_{max,t+2} = \cdots = y_{max,t+H} = 0m \end{cases}$$

The results is shown in Fig. 7. As we can see, the robot can track the spiral trajectory well, and the impedance and position constraints are both ensured. As a conclusion, the proposed method can effectively solve the constrained variable impedance control problem even in complex contact-rich trajectory tracking tasks.

VI. Conclusion

This paper presented QP approaches to design VIC algorithm with both impedance, position and velocity constraints where the novel term "impedance constraints" presented in this paper is significant to ensure the worst contact compliance. To handle the impedance constraints, a novel impedance model where the control input was constructed by impedance-related parameters was established, and we formulated the impedance-constrained VIC design problem as a QP problem. Then, to handle the position and velocity constraints, we introduced the complementary force into the novel impedance model. When the robot will violate the position or velocity constraints, the complementary force will be generated to prevent the violation. The design problem of the complementary force was also transformed as a QP problem. Finally, various experiments were conducted to validate the proposed QP-based VIC algorithm.

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